

Stem-and-Leaf Plots

A **stem-and-leaf plot** displays data by splitting up each value in a dataset into a “stem” and a “leaf.”

How to Make a Stem-and-leaf Plot

The following two examples illustrate how to create a stem-and-leaf plot from scratch for a given dataset.

Example 1

Suppose we have the following dataset:

12, 14, 18, 22, 22, 23, 25, 25, 28, 45, 47, 48

Here is what the stem and leaf plot looks like:

```
1 | 2 4 8
2 | 2 2 3 5 5 8
3 |
4 | 5 7 8
```

And here's how to make it:

Step 1: split each value in the dataset into a “stem” and a “leaf.”

The “**stem**” of each value is the first digit:

12, 14, 18, 22, 22, 23, 25, 25, 28, 45, 47, 48

The “**leaf**” of each value is the second digit:

12, 14, 18, 22, 22, 23, 25, 25, 28, 45, 47, 48

Step 2: To make the plot, place the “stems” along the vertical axis and the “leaves” along the horizontal axis:

```
1 | 2 4 8
2 | 2 2 3 5 5 8
3 |
4 | 5 7 8
```

Example 2

Suppose we have the following dataset:

134, 156, 158, 159, 160, 162, 164

Here is what the stem and leaf plot looks like:

```
13 | 4
14 |
15 | 6 8 9
16 | 0 2 4
```

And here's how to make it:

Step 1: split each value in the dataset into a "stem" and a "leaf."

In this case, since the values in this dataset have three digits, the "stem" of each value is the first two digits:

134, 156, 158, 159, 160, 162, 164

The "leaf" of each value is the last digit:

134, 156, 158, 159, 160, 162, 164

Step 2: To make the plot, place the "stems" along the vertical axis and the "leaves" along the horizontal axis:

```
13 | 4
14 |
15 | 6 8 9
16 | 0 2 4
```

How to Interpret a stem-and-leaf plot

Suppose we have the following stem and leaf plot that shows the number of ice cream cones that Marie sold at her ice cream shop during each of the past 14 days:

```
7 | 3 3 3 7 9
8 | 4 5 8 8
9 | 1 2 5 8 9
```

Question: What is the most cones that Marie sold on any given day?

Answer: 99 cones

7		3	3	3	7	9
8		4	5	8	8	
9		1	2	5	8	9

Question: During how many days did Marie sell more than 80 cones?

Answer: nine days

7		3	3	3	7	9
8		4	5	8	8	
9		1	2	5	8	9

(during these nine days, she sold 84, 85, 88, 88, 91, 92, 95, 98, and 99 cones)